Label Encoding:

Converts categories to numeric labels (0, 1, 2, ...). This can be useful for algorithms that can interpret and deal with ordinal data.

Ordinal data have a meaningful order or ranking, but the intervals between the values are not necessarily equal. For instance, consider a dataset with a column representing education levels:

High School

Bachelor's

Master's

PhD

These education levels have a clear order, making them suitable for label encoding.

In [1]:
```python
import pandas as pd

# Sample data
data = {"name":["alex","deepu","charles","lilly","mary","lekshmi","anju"],'education_level': ['High School',
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
df
```

```
Original DataFrame:
```

Out[1]:

	name	education_level
0	alex	High School
1	deepu	Bachelor
2	charles	Master
3	lilly	PhD
4	mary	Bachelor
5	lekshmi	Master
6	anju	High School

Label Encoding the Ordinal Data

We can use LabelEncoder from sklearn to encode the education levels:

In [2]:
```python


# Define the order for the ordinal data
order = ['High School', 'Bachelor', 'Master', 'PhD']
```

In [3]:

```python
# Create a mapping from the order to numeric values
order_mapping = {label: index for index, label in enumerate(order)}#mention (a number of things) one by one.


# Map the ordinal data to numeric values
df['education_level_encoded'] = df['education_level'].map(order_mapping)

# Display the DataFrame with label encoding
print("\nDataFrame with Label Encoding:")
df
```

```
DataFrame with Label Encoding:
```

Out[3]:

	name	education_level	education_level_encoded
0	alex	High School	0
1	deepu	Bachelor	1
2	charles	Master	2
3	lilly	PhD	3
4	mary	Bachelor	1
5	lekshmi	Master	2
6	anju	High School	0

In [4]:

```python
for index, label in enumerate(order):
    print(index, label)
```

```
0 High School
1 Bachelor
2 Master
3 PhD
```

In [ ]:

In [5]:
```python
# Import label encoder
from sklearn import preprocessing

# label_encoder object knows
# how to understand word labels.
label_encoder = preprocessing.LabelEncoder()

# Encode labels in column 'species'.
df['education_level new']= label_encoder.fit_transform(df['education_level'])

df['education_level'].unique()
```

Out[5]:
```
array(['High School', 'Bachelor', 'Master', 'PhD'], dtype=object)
```

In [6]:
```python
df
```

Out[6]:

	name	education_level	education_level_encoded	education_level new
0	alex	High School	0	1
1	deepu	Bachelor	1	0
2	charles	Master	2	2
3	lilly	PhD	3	3
4	mary	Bachelor	1	0
5	lekshmi	Master	2	2
6	anju	High School	0	1

onehot encode

In [11]:
```
import pandas as pd
import seaborn as sns
iris=sns.load_dataset("iris")


iris
```

Out[11]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

In [13]:
```python
# One-Hot Encode the 'species' column
iris_one_hot_df = pd.get_dummies(iris, columns=['species'])

iris_one_hot_df
```

Out[13]:

	sepal_length	sepal_width	petal_length	petal_width	species_setosa	species_versicolor	species_virginica
0	5.1	3.5	1.4	0.2	1	0	0
1	4.9	3.0	1.4	0.2	1	0	0
2	4.7	3.2	1.3	0.2	1	0	0
3	4.6	3.1	1.5	0.2	1	0	0
4	5.0	3.6	1.4	0.2	1	0	0
...	...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	0	0	1
146	6.3	2.5	5.0	1.9	0	0	1
147	6.5	3.0	5.2	2.0	0	0	1
148	6.2	3.4	5.4	2.3	0	0	1
149	5.9	3.0	5.1	1.8	0	0	1

150 rows × 7 columns

In [ ]: